

TrendScout AI: Conversational Market Intelligence for AI Startups

Powered by RAG + Knowledge Graph

Group: #14 Project: #5 Course: CSE 573 - Semantic Web Mining

Team Members:

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Problem Definition

The Challenge:

- AI startup ecosystem is exploding rapidly
 - 46.4% of US VC funding went to AI in 2024 (up from 36%)
 - \$56 billion raised by generative AI startups in 2024
 - 76% of VCs use AI for daily tasks, 62% use AI to research companies
- Information overload makes manual tracking impractical
 - Funding rounds, product launches, partnerships
 - News articles, LinkedIn posts, announcements
 - Industry narratives evolve quickly (e.g., "AI agents" → "AI infrastructure")

Our Solution:

- TrendScout AI: An "always-on market intelligence analyst"
 - Conversational AI agent that monitors AI startup landscape
 - Combines RAG (semantic search) + Knowledge Graph (structured facts)
- Natural language Q&A about AI startups, products, partnerships

System Architecture

1

Data Flow

Data Collection

- Source: LinkedIn posts from 5 AI companies
- Tool: Selenium + BeautifulSoup web scraping

2

Entity Extraction

- LLM-powered extraction using Gemini API
- Extracts: Companies, Products, Partners, AI Models, Features, Topics

3

Dual Storage Systems

- ChromaDB (Vector Store): For RAG semantic search
- Neo4j (Graph Database): For Knowledge Graph queries

4

Answer Generation

- OpenAI GPT-4 synthesizes answers from both sources
- Combines KG structured facts + RAG contextual details

5

User Interface

- Flask web application (index.html)
- Toggle KG on/off, RAG on/off
- Toggle source context

Algorithms & Methods

RAG Pipeline

(trendscout_app.py)

- Text chunking: RecursiveCharacterTextSplitter
- Embeddings: OpenAI text-embedding-ada-002
- Vector store: ChromaDB with cosine similarity search
- Smart source selection: 1-5 sources based on keywords in the query
 - Factual queries (“what”, “who”, “when”) → 2 sources
 - Comparison queries (“compare”, “difference”) → 4 sources
 - List/overview queries (“list”, “all”, “summarize”) → 5 sources
- Score threshold filtering for relevance

Hybrid Query Approach:

User query → KG retrieval (structured) + RAG retrieval (semantic) → LLM

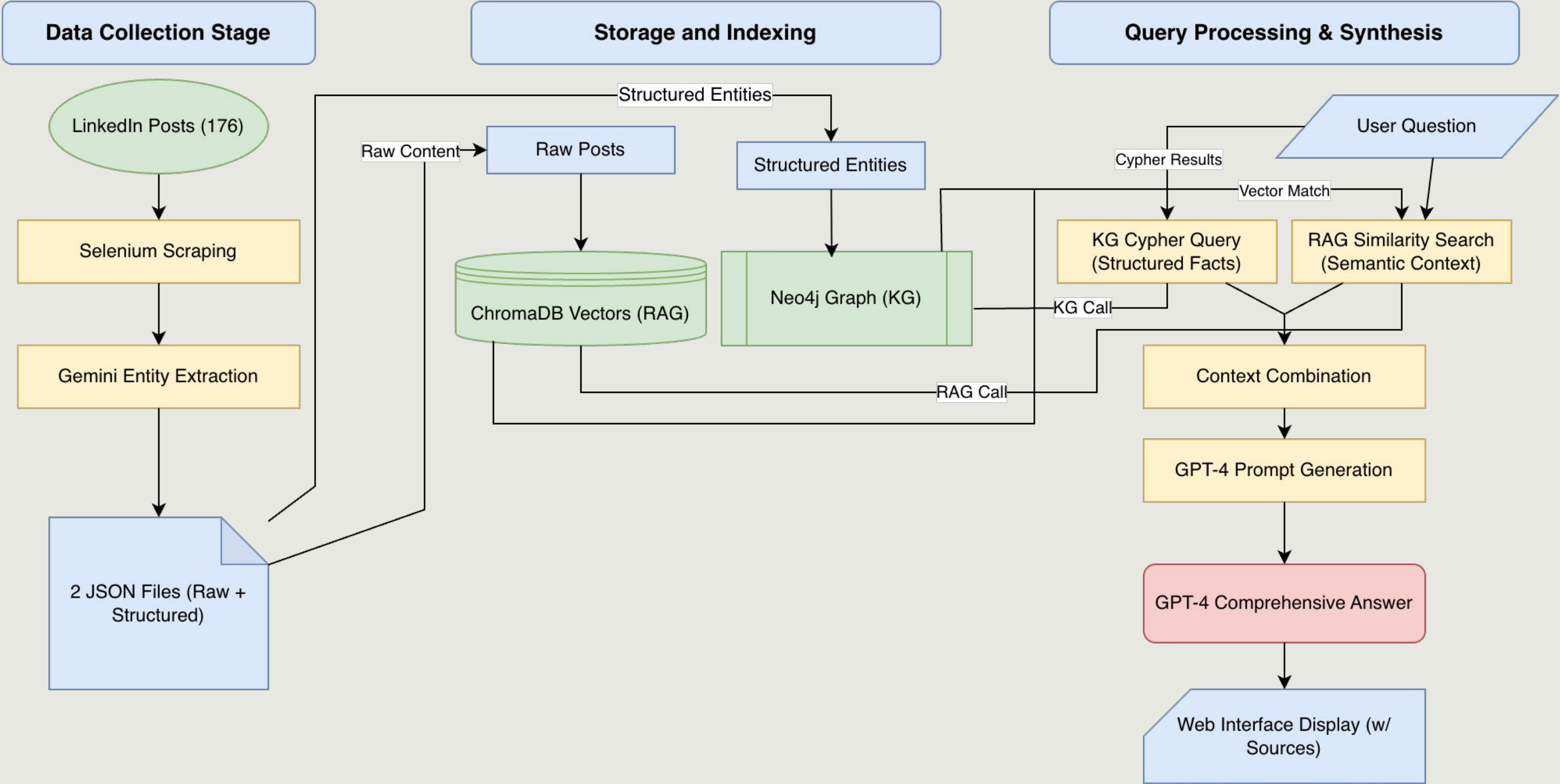
→ Answer

Knowledge Graph

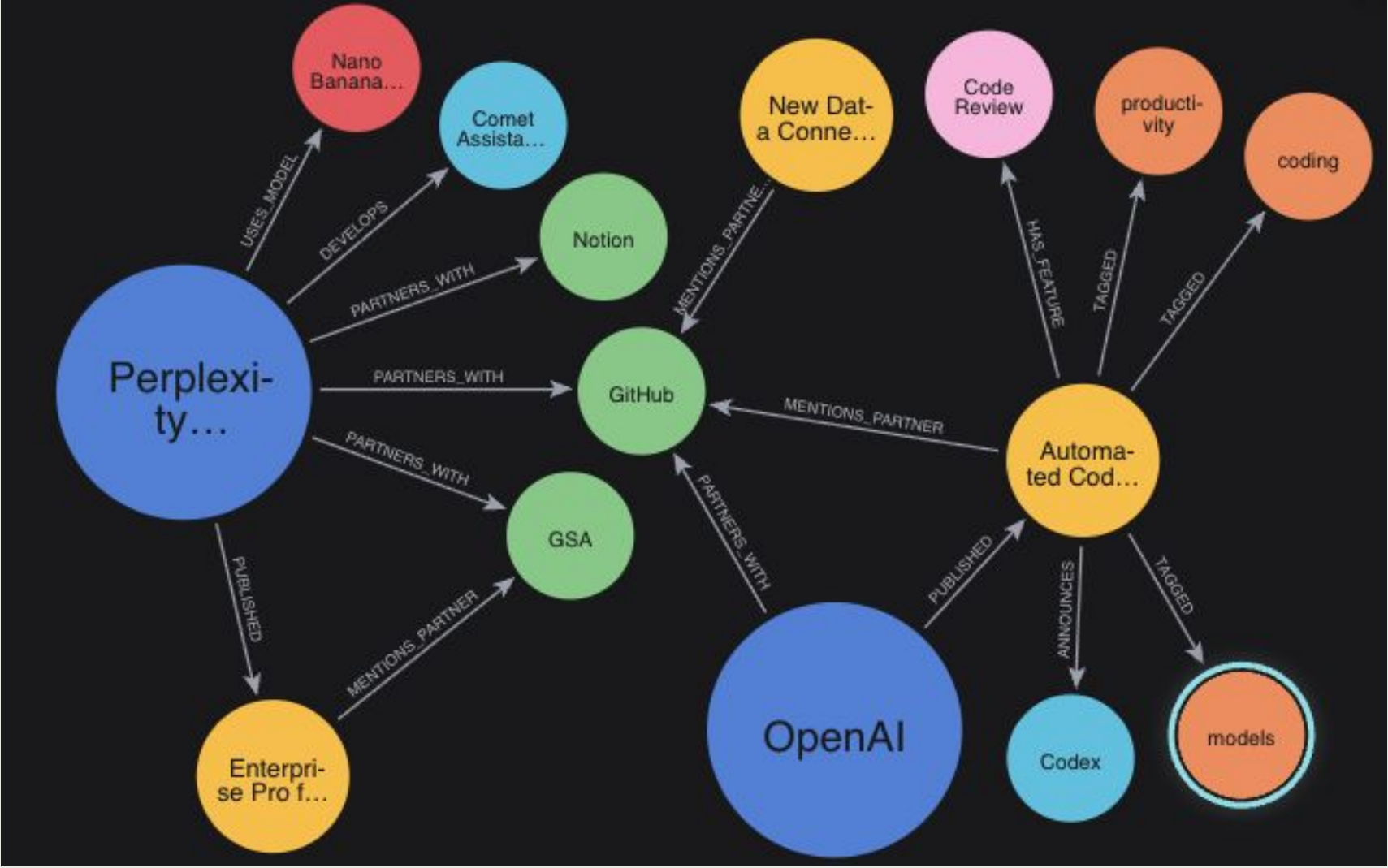
(load_linkedin_posts_kg.py)

- Entity extraction: LLM-powered (Gemini API) - more accurate than regex
- Graph database: Neo4j with Cypher query language
- Node types: Company, Product, Partner, AIModel, Feature, Topic, Post
- Relationship types:
 - Company → DEVELOPS → Product
 - Company → PARTNERS_WITH → Partner
 - Company → SUPPORTS → AIModel
 - Post → MENTIONS → Product/AIModel
 - Post → TAGGED → Topic

DATA FLOW



Neo4j Graph Visualization:



Node types:

- Company, Product, Partner
- Post, AIModel, Feature, Topic

Relationship types:

- Company → DEVELOPS → Product
- Company → PARTNERS_WITH → Partner
- Company → SUPPORTS → AIModel
- Company → PUBLISHED → Post
- Post → METIONS_PARTNER → Partner
- Post → MENTIONS → Product/AIModel
- Post → TAGGED → Topic
- Post → ANNOUNCES → Product

Datasets

Data Source:

LinkedIn company posts from 5 leading AI startups

Dataset Size:

Company	Posts
Perplexity AI	40
OpenAI	40
Anthropic	40
Mistral AI	40
DeepSeek	16
Total	176 posts

Date Range: March 2025 - November 2025

Preprocessing Steps:

- Web scraping with Selenium + BeautifulSoup
- LLM-powered entity extraction (Gemini API)
- Convert to structured JSON format
- Load into ChromaDB (vectors) and Neo4j (graph)

Knowledge Graph Statistics:

- 5 Companies
- 176 Posts
- 144 Products
- 63 AI Models
- 148 Partners
- Multiple Topics and Features

Evaluation Methodology

Goal: Compare RAG-only vs RAG+KG system performance

Test Set:

50 evaluation questions across 5 categories:

- Factual (10): "What products does Anthropic offer?"
- Relationship (10): "Which companies partner with government agencies?"
- Comparison (10): "Compare Perplexity vs OpenAI features"
- Temporal (10): "When did Mistral announce their API?"
- Detail (10): "Describe Claude's coding capabilities"

Scoring Method:

Automated evaluation using Gemini API (flash 2.0)

Three metrics (each 1-5 scale):

- Accuracy: Does answer match expected? (5)
- Completeness: Does it cover all relevant information? (5)
- Relevance: Is it focused on the question? (5)

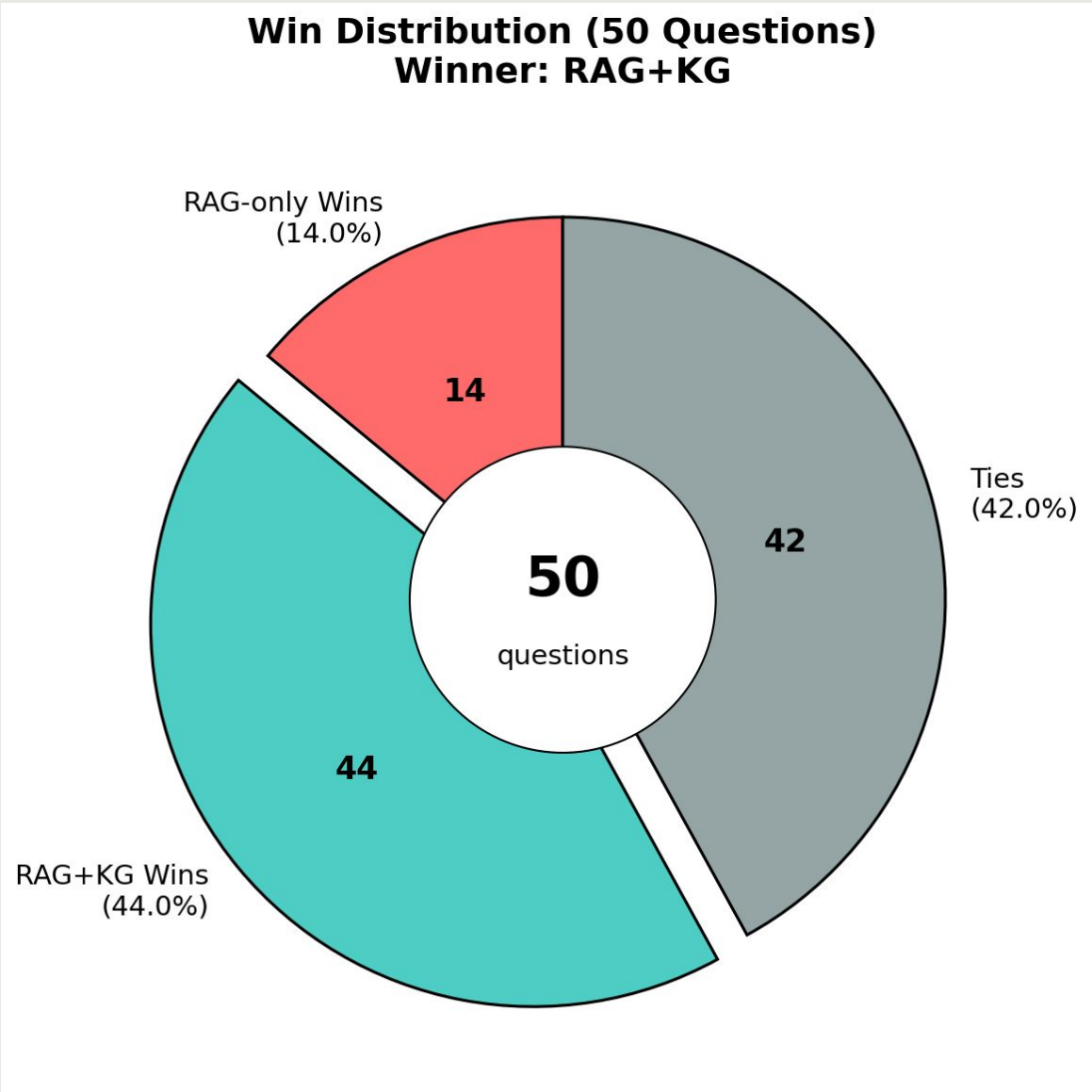
Total score: 15 per answer

Evaluation Pipeline:

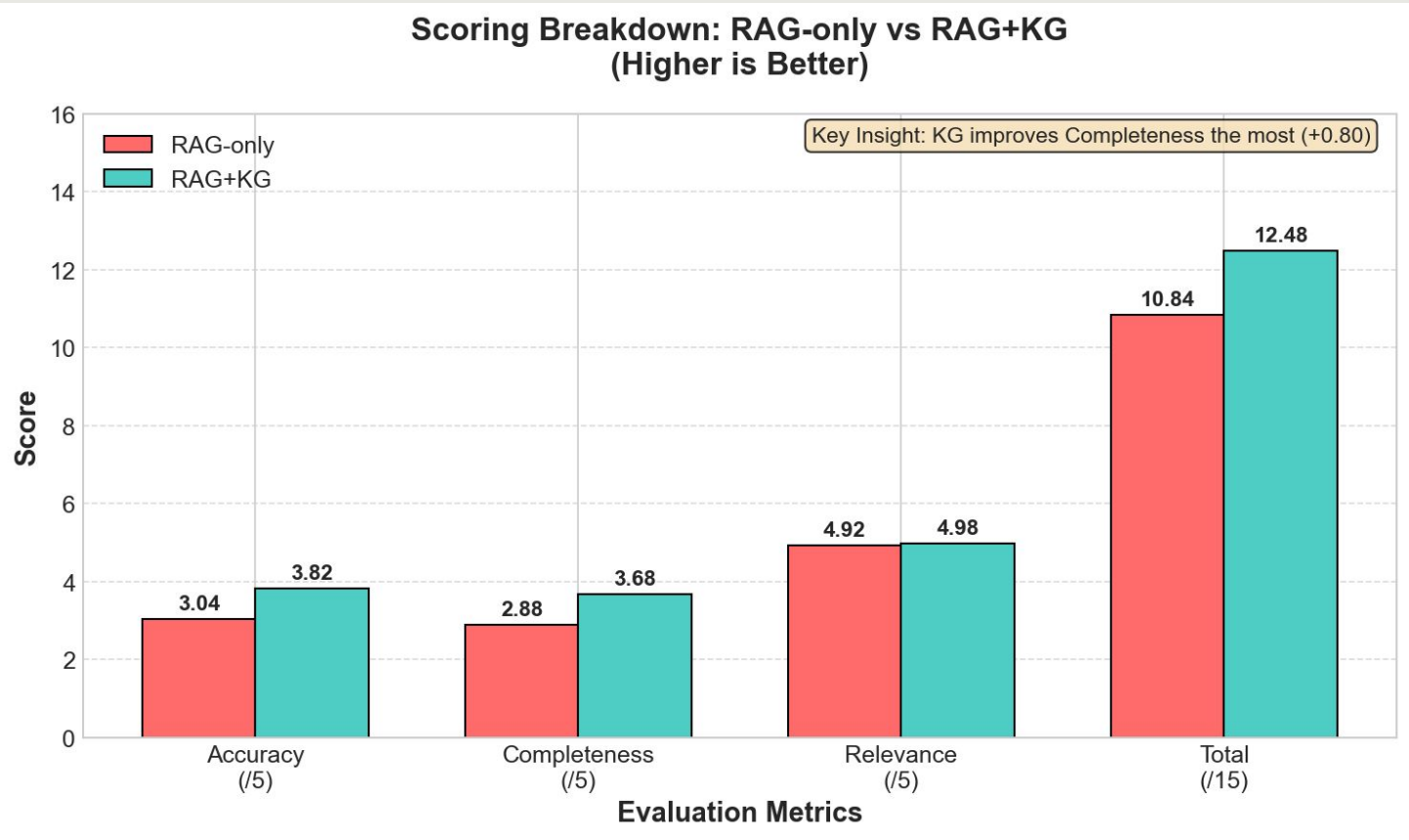
1. Load 50 questions with expected answers
2. Run each question through RAG-only system
3. Run each question through RAG+KG system
4. Send both answers to Gemini for scoring
5. Summary comparison report

Evaluation Results

Overall Results (50 Questions):

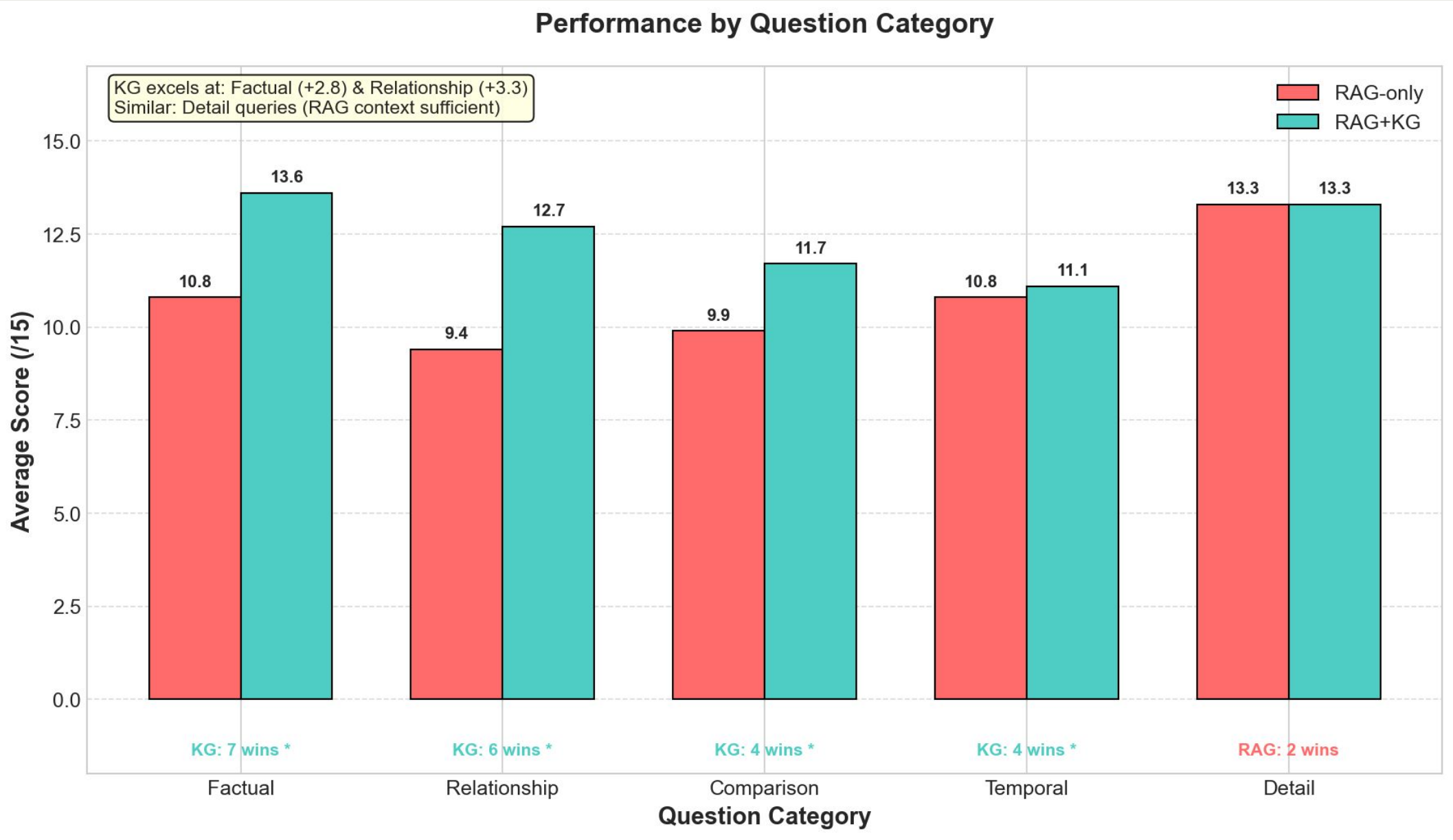


Scoring Breakdown (Averages):



Evaluation Results

Performance by Question Category:



DEMO



AI Startup Intelligence powered by RAG + Knowledge Graph

- "Who are the investors in NestAI?"
- "Show me AI startups in Berlin"

What products has Perplexity AI made?

TrendScout AI  KG  RAG

Perplexity AI has developed several products, including:

1. **Perplexity** - A default AI for Snapchat.
2. **Comet** - A powerful personal AI assistant.
3. **Email Assistant** - A private email drafting and labeling tool.
4. **Perplexity Patents** - The world's first AI patent research agent, designed to make IP intelligence accessible.
5. **Comet Assistant** - An updated version of Comet with multi-site workflow capabilities.
6. **Enterprise Pro for Government** - A powerful, model-agnostic AI platform with secure-by-default guarantees for federal users.

These products cater to various needs, from personal assistance to specialized applications in intellectual property and government services.

 Show/Hide Sources

Ask

✓ Use Knowledge Graph ✓ Use RAG

System Status: ● KG Connected | ● RAG Ready

Project Plan & GitHub

Division of Work:

Team Member	Responsibilities
Aditya Pokharna	AI Integration and Orchestration: Focusing on integrating Transformer models and overall system workflow management.
Sahil Pawar	Chatbot Logic and LangFlow Integration: Designing the conversational flow and implementing the RAG pipeline using LangChain/LangFlow.
Wei-An Wang	Core System Development: Entity extraction (Gemini), Knowledge Graph, RAG pipeline (LangFlow & LangChain), Flask web app, Evaluation, DEMO
Yu-Yao Hsieh	Data Collection and Baseline ML: Implementing web scraping routines (Selenium) on various website(techfundingnews, venturebeat, YC jobs).
Zih-Jyun Lin	Conversational Design and Data Collection: Defining user interaction patterns and developing the quantitative/qualitative evaluation framework. Linkedin post scraping.

GitHub Repository: [GitHub - jason7580/CSE573-TrendScout-Lite](https://github.com/jason7580/CSE573-TrendScout-Lite)

Thanks for watching